UVM Verification of a

UART Core

Results

by

Hossam Shaher Anis

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Contents

Results 1
Contents 2
List of Tables 3
1 Introduction 4
2 Testing Results 5
3 Coverage Results 6
4 Assertion Results 7

List of Tables

Table 1 Coverage results 6

1 Introduction

Serial communication uses a single data line (***serial line***) to exchange information between two systems.

The transmitting system converts the parallel data to a serial stream, and the receiving system reassembles the serial data back to its original parallel format.

A **UART (Universal Asynchronous Receiver and Transmitter) core** is the most commonly used scheme.

In this project, the UART core is verified using the SystemVerilog HDVL and the Universal Verification Methodology (UVM).

A verification environment was developed[1], and a verification plan (test plan, coverage plan, and assertion (checking) plan) was set[2].

This document contains:

1. Testing results.
2. Coverage results.
3. Assertion (checking) results.

[1] See "*Data Aligner - Architecture of The Verification Environment*".
[2] See "*Data Aligner – Verification Plan*".

2 Testing Results

The DUT successfully passed all test cases.

3 Coverage Results

Table 1 shows coverage results.

Table 1 Coverage results

#	Coverage Collector[3]	`covergroup/ cover property`	`coverpoint/ cross/ property`	Coverage (%)	Test and Seed[4]
1	m_uart_coverage_collector_system	cover_group_system	dvsr	100.00%	uart_test_dynamic_divisor (1)
			reset	100.00%	uart_test_reset (1)
2	m_uart_coverage_collector_rx	cover_group_rx	dout	100.00%	uart_test_directed_data (1)
			dvsr	100.00%	uart_test_dynamic_divisor (1)
			reset	100.00%	uart_test_reset (1)
3	m_uart_coverage_collector_tx	cover_group_tx	tx	100.00%	uart_test_directed_data (1)
			dvsr	100.00%	uart_test_dynamic_divisor (1)
			reset	100.00%	uart_test_reset (1)
4	uart_rx_if_inst	UART_FRAME_RX_C	uart_frame_rx_p	8056 attempts, 4 match	uart_test_normal_operation (1)
5	uart_tx_if_inst	UART_FRAME_TX_C	uart_frame_tx_p	8056 attempts, 4 match	uart_test_normal_operation (1)

[3] Coverage collector = Class instance or interface instance

[4] Tests and seeds used to obtain these results. The seed is enclosed in round brackets.

4 Assertion Results

The DUT succeeded in all assertions. No errors were reported.